

Bharat Badgujar

Linux module 2

Assignment 2

1.

Install ufw on your Linux system (if not already installed) and enable it, then verify its status using the appropriate commands.

Ans : **Commands**

```
sudo apt update
```

```
sudo apt install ufw -y
```

```
sudo ufw enable
```

```
sudo ufw status verbose
```

Expected Output

Status: active

2.

Configure ufw to allow incoming SSH connections only on port 22 and block all other incoming connections.

Ans: **Commands**

```
sudo ufw default deny incoming
```

```
sudo ufw allow 22/tcp
```

```
sudo ufw reload
```

```
sudo ufw status numbered
```

Expected Output

```
22/tcp ALLOW IN Anywhere
```

This configuration allows SSH access on **port 22** while blocking all other incoming connections.

3.

Add a rule in ufw to allow outgoing HTTP and HTTPS traffic (ports 80 and 443), but block all other outbound connections.

Hint: Use 'ufw allow out' and 'ufw deny out' commands to control outbound traffic.

Ans : **Commands**

```
sudo ufw default deny outgoing
```

```
sudo ufw allow out 80/tcp
```

```
sudo ufw allow out 443/tcp
```

```
sudo ufw reload
```

```
sudo ufw status verbose
```

Expected Output

```
80/tcp ALLOW OUT Anywhere
```

```
443/tcp ALLOW OUT Anywhere
```

This configuration allows only **HTTP (80)** and **HTTPS (443)** outbound traffic while blocking all other outgoing connections.

4.

List all currently active ufw rules on your system and save the output to a file named firewall_rules.txt.

Ans : **Command**

```
sudo ufw status numbered > firewall_rules.txt
```

Verify the File

```
cat firewall_rules.txt
```

5.

Write a short explanation (3-4 lines) describing how your current ufw configuration would impact running a local web server (like a development server for a Zomato-style food delivery app).

Ans : With this UFW configuration, only **SSH (port 22)** is allowed for incoming connections, so users cannot access the local web server because ports like **80** or **8080** are blocked. Outgoing traffic is limited to **HTTP (80)** and **HTTPS (443)**, allowing the server to download updates or access web APIs. To make the web application accessible from other systems, the required web server port (such as **80**, **443**, or **8080**) must be explicitly allowed in UFW.